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d) Geometric Random Variable

$X \sim Geo_1(p)$, $p \in (0,1]$

$P_X(x) = p(1-p)^{x-1}$, $x \in \mathbb{N}$

: number of independent Bernoulli trials needed until the first success is observed

$X \sim Geo_0(p)$, $P_X(x) = p(1-p)^x$ $x \in \mathbb{Z}^{\geq 0}$

Exercise: $X \sim Geo_0(p)$ or $Y \sim Geo_1(p)$

show $\sum_{x=0}^{\infty} P_X(x) = 1$ or $\sum_{x=1}^{\infty} P_Y(y) = 1$

i) $X \sim Geo_0(p)$

$P_X(x) = p(1-p)^x$

$\Rightarrow \sum_{x=0}^{\infty} p(1-p)^x = \frac{p}{1-(1-p)} = 1$

ii) $Y \sim Geo_1(p)$

$P_Y(y) = p(1-p)^{y-1}$

$\Rightarrow \sum_{y=1}^{\infty} p(1-p)^{y-1} = \frac{p}{1-(1-p)} = 1$

e) Poisson Random Variable

$X \sim Pois(\lambda)$

$P_X(x) = \frac{\lambda^x}{x!} e^{-\lambda}$, $x = 0, 1, 2, \dots$

$\sum_{x=0}^{\infty} P_X(x) = \sum_{x=0}^{\infty} e^{-\lambda} \frac{\lambda^x}{x!} = e^{-\lambda} \cdot e^{\lambda} = 1$

Poisson Distribution as a limit of the binomial

$\lambda = np$

$$\frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \approx e^{-\lambda} \frac{\lambda^x}{x!}$$

assume $X_n \sim Bin(n, \frac{\lambda}{n})$

$$\lim_{n\to\infty} P_{X_n}(x) = \lim_{n\to\infty} \frac{n!}{x!(n-x)!} \left(\frac{\lambda}{n}\right)^x \left(1-\frac{\lambda}{n}\right)^{n-x}$$

$$= \lim_{n\to\infty} \frac{\lambda^x}{x!} \cdot \frac{n!}{(n-x)!\,n^x} \cdot \left(1-\frac{\lambda}{n}\right)^n \left(1-\frac{\lambda}{n}\right)^{-x}$$

$$= \lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^n \cdot \frac{\lambda^x}{x!} = e^{-\lambda} \cdot \frac{\lambda^x}{x!}$$

f) Negative Binomial

$X \sim NegB(k, p)$; number of independent Bernoulli trials until success is seen k times

$$P_X(x) = \binom{x-1}{k-1} p^{k-1} (1-p)^{x-k} \cdot p$$

$$= \binom{x-1}{k-1} p^k (1-p)^{x-k} \quad , \; x \geq 1 \, , \; x \geq k$$

g) Power Law Distribution

i) Zipf - Mandelbrot Distribution

$$P_X(x) = \frac{c}{(x+q)^\alpha} \qquad x = 1, 2, \cdots, N$$

$$\sum_{x\in\mathbb{N}} P_X(x) = 1 = \sum \frac{c}{(x+q)^\alpha}$$

$$c = 1 \Big/ \sum \frac{1}{(x+q)^\alpha}$$

ii) Hurwitz Zeta function $\zeta(\alpha, q)$

$$P_X(x) = \frac{c}{(x+q)^\alpha}$$

iii) Another Power Law

$$P_X(x) = \frac{1}{x^\alpha} - \frac{1}{(x+1)^\alpha} \quad x \in \mathbb{N}, \ \alpha > 0$$

Exercise : Show $\sum_{x \in \mathbb{N}} P_X(x) = 1$

$$S_1 = \frac{1}{1^\alpha} - \frac{1}{2^\alpha}$$

$$S_2 = \frac{1}{1^\alpha} - \frac{1}{2^\alpha} + \frac{1}{2^\alpha} - \frac{1}{3^\alpha}$$

$$\vdots$$

$$\therefore \lim_{n \to \infty} S_n = \sum_{x \in \mathbb{N}} P_X(x) = 1 - \lim_{n \to \infty} \frac{1}{n^\alpha} = 1 \qquad \forall \alpha > 0$$